ReLU Site-scope Pressure Comparison

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1 Immediate Question

This report compares only the completed ReLU one-pass MiniPile runs. Pythia-14M is used as a random-initialized architecture with the MLP activation switched to ReLU; no released Pythia checkpoint weights are loaded.

> Does ReLU keep the high MLP-hidden sparsity regime when pressure is moved from MLP-only to MLP+residual and all-site pressure, and where does the near-zero mass appear across activation families?

The AdamW ReLU run, config 77, is the baseline for all comparisons. The pressure settings are fixed across scopes: RN and OR use $w = 1, c = 0.05, \sigma = 0.05$; L1N and OL1 use $w = 5$. Site scopes are:

- **MLP-only**: pressure on `mlp_hiddens`; configs 78–81.
- **MLP+residual**: pressure on `mlp_hiddens` and `residual_streams`; configs 86–89.
- **All-site**: pressure on `mlp_hiddens`, `residual_streams`, and `attention_outputs`; configs 90–93.

Table 1: Endpoint summary for completed ReLU site-scope runs. Validation loss and logged near-zero mass come from each run's `metrics.json`. The MLP, residual, and attention columns are validation-cache histogram estimates, averaged over layers 0–5.

Scope	Method	Config	Val. loss	Δ vs AdamW	Logged NZ	MLP NZ	Resid. NZ	Attn NZ
Baseline	AdamW	77	4.8404	+0.0000	53.9	53.6	13.3	51.4
MLP-only	RN	78	4.8134	-0.0271	94.2	94.0	16.0	43.9
MLP-only	OR	79	4.8087	-0.0317	93.6	93.5	16.4	50.1
MLP-only	L1N	80	4.7967	-0.0437	95.8	96.1	16.4	37.8
MLP-only	OL1	81	4.7981	-0.0423	93.6	94.1	18.4	47.5
MLP+residual	RN	86	5.0495	+0.2091	74.9	83.3	30.5	41.8
MLP+residual	OR	87	4.9629	+0.1224	78.1	85.7	35.8	54.9
MLP+residual	L1N	88	4.8233	-0.0172	79.0	90.5	30.8	73.2
MLP+residual	OL1	89	4.8293	-0.0112	77.3	88.9	30.3	72.1
All-site	RN	90	5.0024	+0.1619	78.2	84.2	33.5	97.4
All-site	OR	91	4.8903	+0.0498	78.5	84.7	36.2	95.7
All-site	L1N	92	4.7492	-0.0913	73.8	87.1	27.8	85.0
All-site	OL1	93	4.8263	-0.0142	74.3	85.9	27.8	90.3

Readout. MLP-only ReLU pressure keeps the cleanest high-MLP-sparsity regime: about 94–96% MLP near-zero mass with validation loss slightly below the ReLU AdamW point estimate. Adding residual pressure reduces the aggregate logged sparsity and separates methods more strongly: L1N and OL1 stay near AdamW validation loss, while RN/OR pay a larger loss cost. All-site pressure moves attention outputs into the near-saturated near-zero regime; L1N all-site is the best single point estimate here, but this is still one seed.

2 Learning Curves

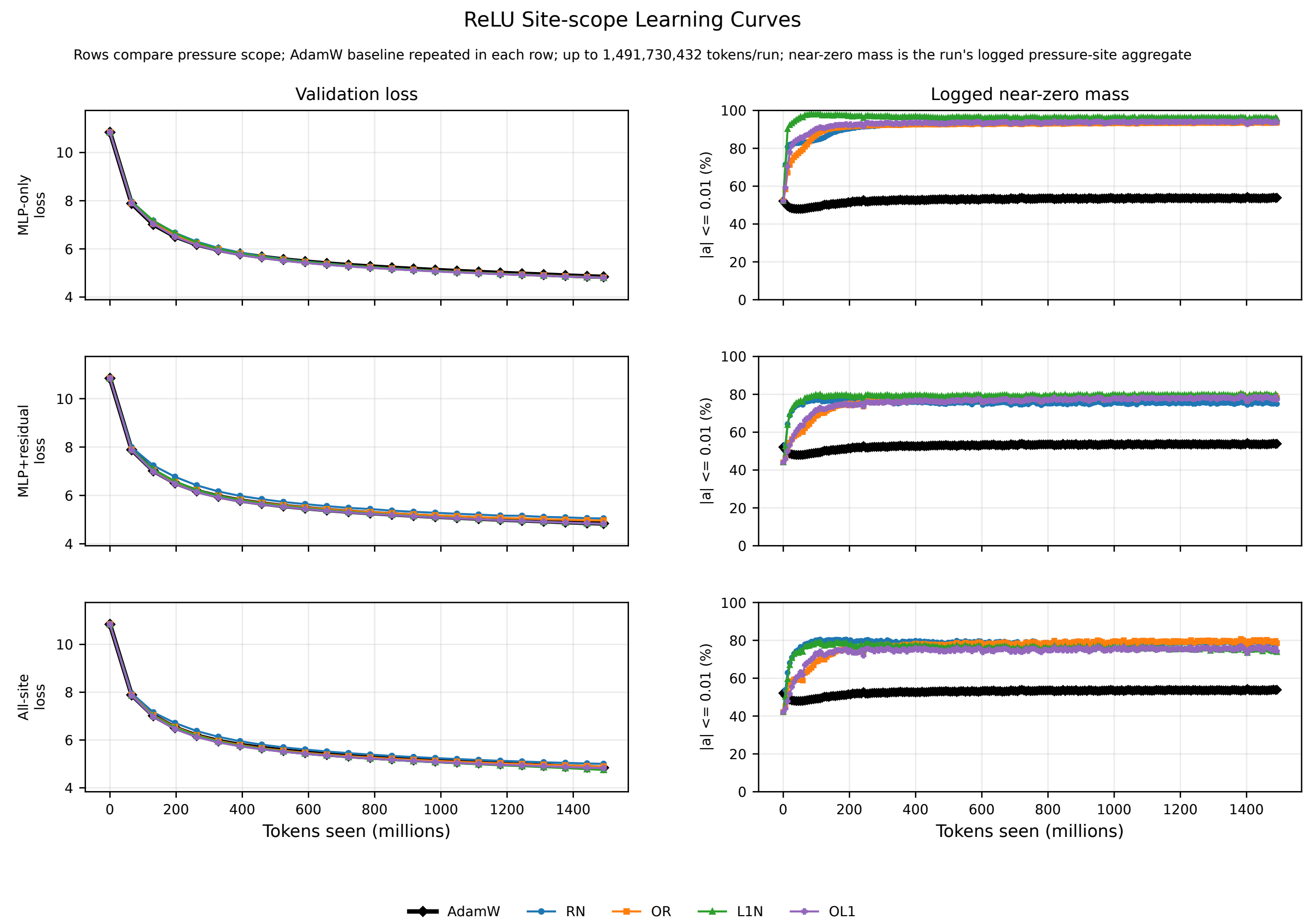

Figure 1: ReLU site-scope learning curves. Rows compare MLP-only, MLP+residual, and all-site pressure; AdamW is repeated as the baseline in each row. The near-zero panel uses each run's logged pressure-site aggregate.

3 Per-layer Near-zero Mass

The heatmap and table below use the validation-cache histogram payloads for configs 94–97. Values are histogram-estimated $|a| \leq 0.01$ percentages by layer [0–5], so they should be read as diagnostic distributions rather than exact training metrics.

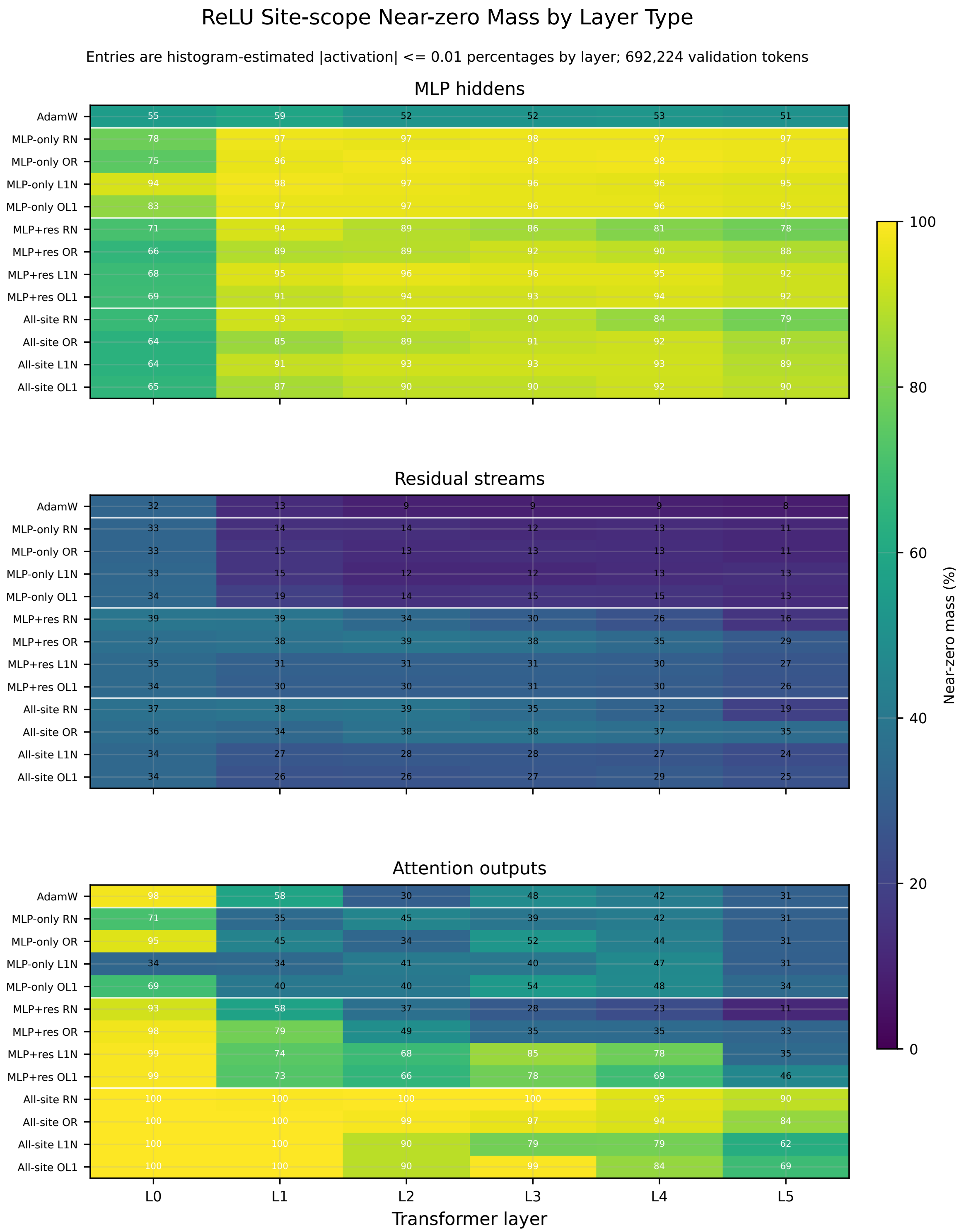

Figure 2: Per-layer near-zero mass by activation family for the ReLU site-scope comparison. MLP-only pressure is strongest on MLP hiddens; all-site pressure shifts the attention-output rows toward saturation.

Table 2: Validation-cache near-zero mass vectors for ReLU site-scope runs. Entries are histogram-estimated $|a| \le 0.01$ percentages by layer [0–5].

Run	MLP hiddens	Residual streams	Attention outputs
Baseline AdamW	[55, 59, 52, 52, 53, 51]	[32, 13, 9, 9, 9, 8]	[98, 58, 30, 48, 42, 31]
MLP-only RN	[78, 97, 97, 98, 97, 97]	[33, 14, 14, 12, 13, 11]	[71, 35, 45, 39, 42, 31]
MLP-only OR	[75, 96, 98, 98, 98, 97]	[33, 15, 13, 13, 13, 11]	[95, 45, 34, 52, 44, 31]
MLP-only L1N	[94, 98, 97, 96, 96, 95]	[33, 15, 12, 12, 13, 13]	[34, 34, 41, 40, 47, 31]
MLP-only OL1	[83, 97, 97, 96, 96, 95]	[34, 19, 14, 15, 15, 13]	[69, 40, 40, 54, 48, 34]
MLP+residual RN	[71, 94, 89, 86, 81, 78]	[39, 39, 34, 30, 26, 16]	[93, 58, 37, 28, 23, 11]
MLP+residual OR	[66, 89, 89, 92, 90, 88]	[37, 38, 39, 38, 35, 29]	[98, 79, 49, 35, 35, 33]
MLP+residual L1N	[68, 95, 96, 96, 95, 92]	[35, 31, 31, 31, 30, 27]	[99, 74, 68, 85, 78, 35]
MLP+residual OL1	[69, 91, 94, 93, 94, 92]	[34, 30, 30, 31, 30, 26]	[99, 73, 66, 78, 69, 46]
All-site RN	[67, 93, 92, 90, 84, 79]	[37, 38, 39, 35, 32, 19]	[100, 100, 100, 100, 95, 90]
All-site OR	[64, 85, 89, 91, 92, 87]	[36, 34, 38, 38, 37, 35]	[100, 100, 99, 97, 94, 84]
All-site L1N	[64, 91, 93, 93, 93, 89]	[34, 27, 28, 28, 27, 24]	[100, 100, 90, 79, 79, 62]
All-site OL1	[65, 87, 90, 90, 92, 90]	[34, 26, 26, 27, 29, 25]	[100, 100, 90, 99, 84, 69]

4 Post-hoc Clipping Frontiers

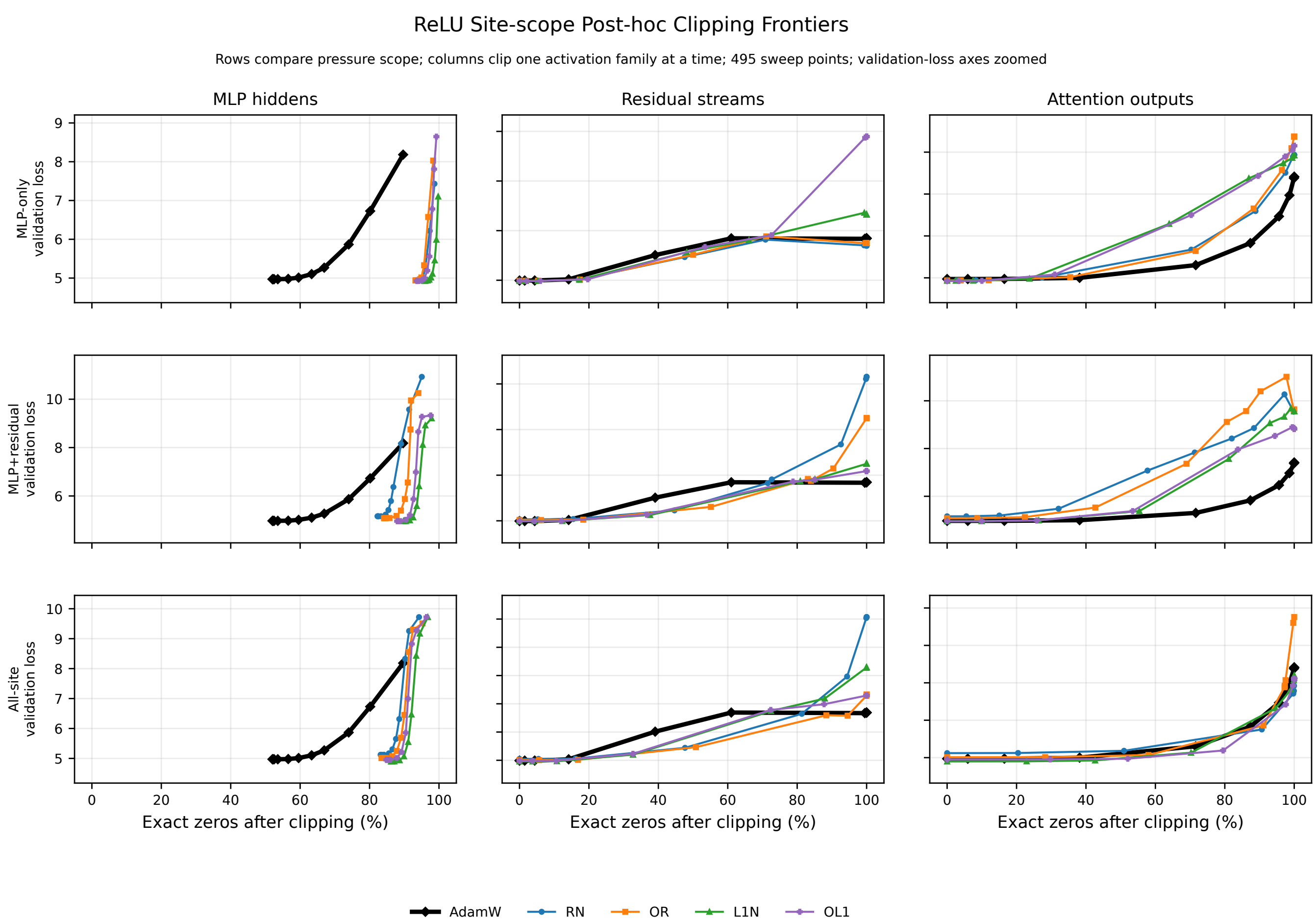

Figure 3: ReLU site-scope post-hoc clipping frontiers. Each column clips only one activation family. MLP-hidden clipping remains the key direct compute-skip proxy; residual and attention clipping show where pressure has moved robustness or fragility off the MLP path.

5 Activation Densities

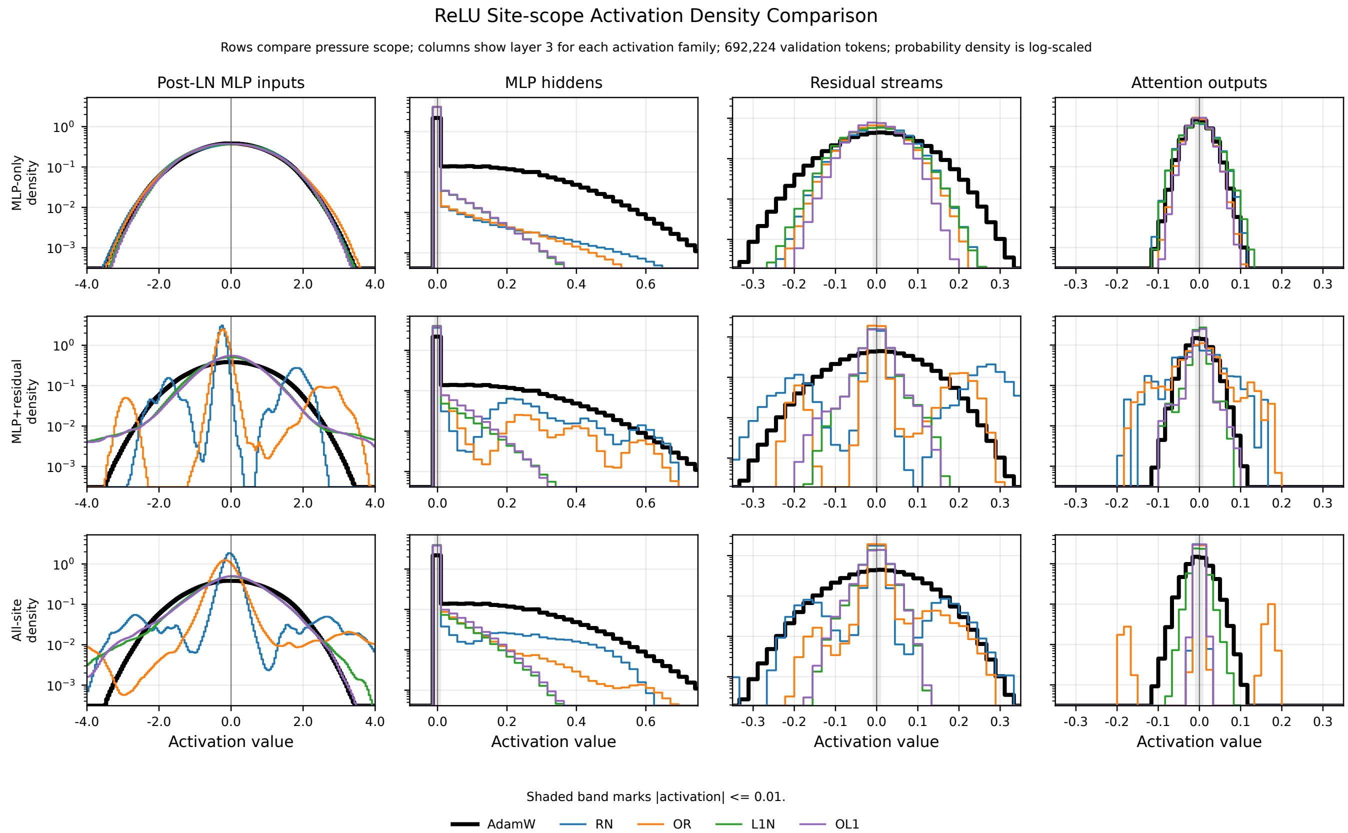

Figure 4: ReLU activation-density comparison at layer 3. The post-LayerNorm MLP-input column exposes how residual-side pressure reshapes inputs before the ReLU MLP hidden site; probability density is log-scaled.

6 Weight Densities

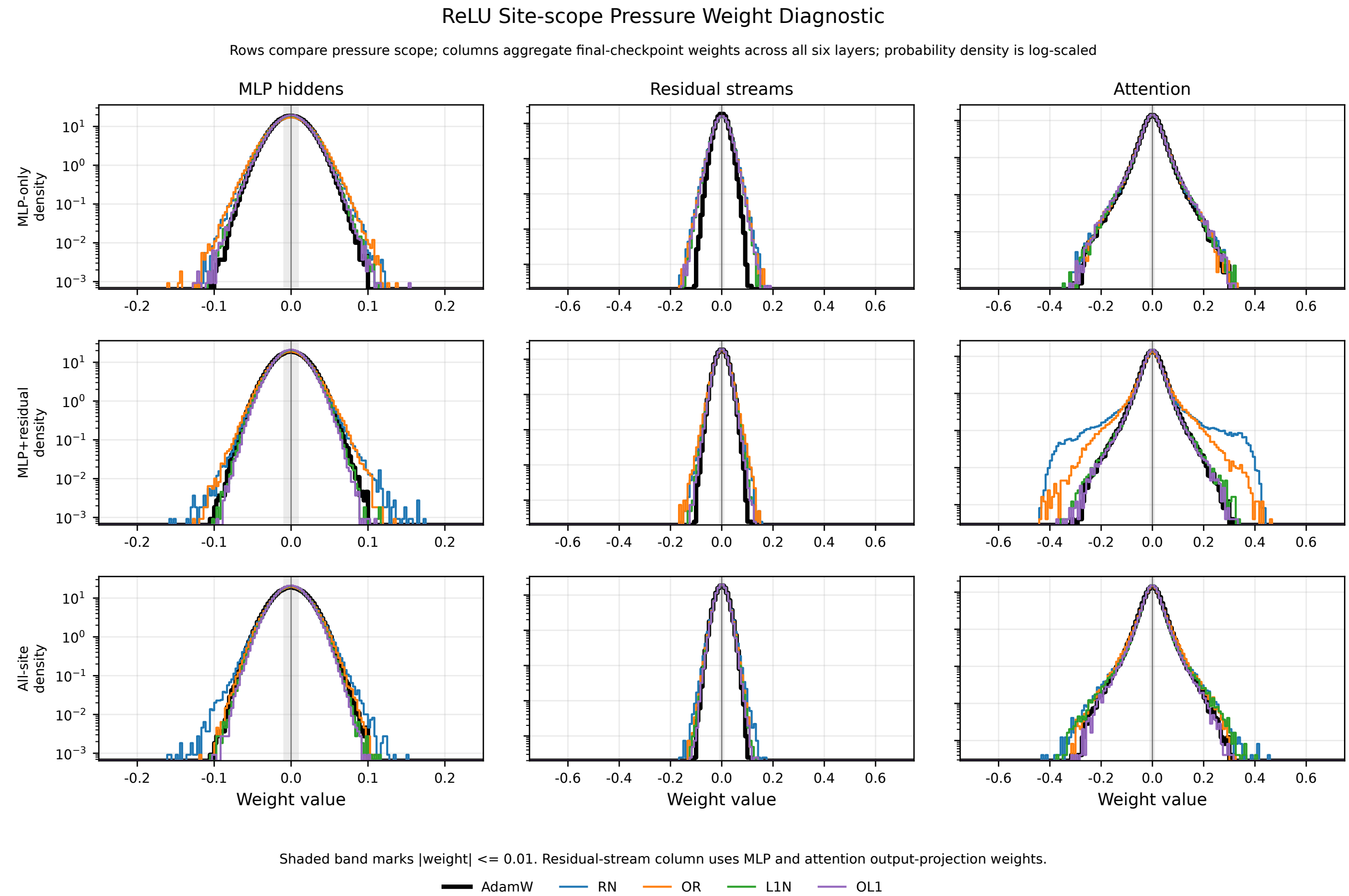

Figure 5: ReLU pressure-weight diagnostic. Columns aggregate final-checkpoint weights over MLP hidden-input weights, output-projection weights that write to residual streams, and attention QKV weights. Weight densities do not imply structured parameter skipping by themselves; they show whether activation pressure is visibly reshaping the corresponding parameter families.

7 Caveats

- These are one-seed runs. Treat differences below a few hundredths of validation loss as planning evidence, not a final claim.
- The clipping sweeps use 8 validation batches per threshold. The activation-density and near-zero tables use the full deterministic validation cache used by the histogram configs.
- The near-zero vectors are histogram-estimated because the histogram artifacts store binned counts, not exact threshold counters for every off-target site.
- Activation sparsity is elementwise. Turning it into skipped computation still requires a structured execution rule, especially for upstream residual and MLP up-projection work.